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Github Repo:- github.com/Thegreatkk04/cloud-assignment

When I started the project, I thought the main work would be training the model I soon realised there was much more around it. I need to collect all the answer for the given questions, check them, train the smaller model, store it and then work through an API.

I also learn a lot from data cleaning part. The main idea was knowledge distillation on the cloud side I learn how the object storage fits into deployment. The model weights are kept in seaweedfs through its S3 API not inside the docker image or in git.

I also learnt the purpose of each Kubernetes object Deploy for replicas and rolling updates service for a stable address ingress for routing/paths, HPA for autoscaling on CPU

finally I also understood what a CI quality gate is The pipeline runs lint, tests and then score.py with --min validity 95, so a bad model fails the build automatically.

2. challenge & faced :-

The biggest challenge was time, labelling 3,000 commands with a 3.8B model on a laptop with CPU take hours and training also takes long. & started and wait for so long to finish it

Two of unit test failed at first. One compared floating point values too strictly. The other expected a packet to be denied but when & read the NACL rules again & saw packet was apply to traffic. The tests were wrong not the code.

Keeping measured result and assumption was also harder than & expected because it is tempting to fill every table

3. How & handled them

& run everything the NACL, the Subnet table, 13 unit test passing, lint with no errors the untrained by this - 70% scored on the validation set and the cost formula

& fixed the two tests after checking the rules and did not change the code to make the test pass

4. finding and observations:-

The untrained python- Tom gave 0% valid lesson on the 100 validation commands & produced random text such as name: "Calendar" or a single { This shows clearly why the training is required and what CI gate will catch.

The first CI run on github failed in two places: the evaluation setup because the secret scan action because of a problem with the first commit & fixed the flow so that the gate still runs on the base model and gitleaks runs directly.

5. What could be done better:-

A 50-record training made test early would have proved the pipeline. Building the docker image early would have confirmed the 2GB limit. Running the Github workflow once with the base model early would have given the failed gate link without stress.

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2. Challenge I faced :-

6. Improvement for next time :-

Use a weak queue for labelling instead of static shards. Use more epochs or a bigger student if full frame accuracy is low. Cache the model in a volume so pods are not blocked when storage is down. Measure real throughput and scale the cost estimate with the measured value.